



**UNIVERSITÉ DE
FRANCHE-COMTÉ**

**PRACTICAL WORK "ASTROPHYSICAL SPECTROSCOPY"
MASTER 2 COMPUPHYS**

**Spectroscopic analysis of planetary atmospheres
and surfaces**

Ali FADLALLAH

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1 Introduction

In this report, we explore the radicals OH found in the Earth's atmosphere at 87 km above the surface. These radicals emit near-infrared light, and we want to study the rotational-vibrational structure of the OH bands, particularly within the $\Delta\nu=2$ to 5 sequences.

We use laboratory data on energy levels and transitions, and created a python code the emission spectrum. The objective is to replicate the observed spectrum, considering a rotational temperature (T_{rot}) of 200 K and a vibrational temperature (T_{vib}) of 9000 K.

2 Spectrum Overview

We run the Jupyter script "main.ipynb" to produce a synthetic spectrum that includes emission lines for the $\Delta\nu=2$ to 5 sequences. Using $T_{rot}=200$ K, $T_{vib}=9000$ K, and $N=3\times 10^{10}$ molecules/cm², the intensities of these emission lines were calculated.

The resulting spectrum, shown in the plot below (Figure 1), showcases all the emission lines for both substates $X^2\Pi_{3/2}$ and $X^2\Pi_{1/2}$.

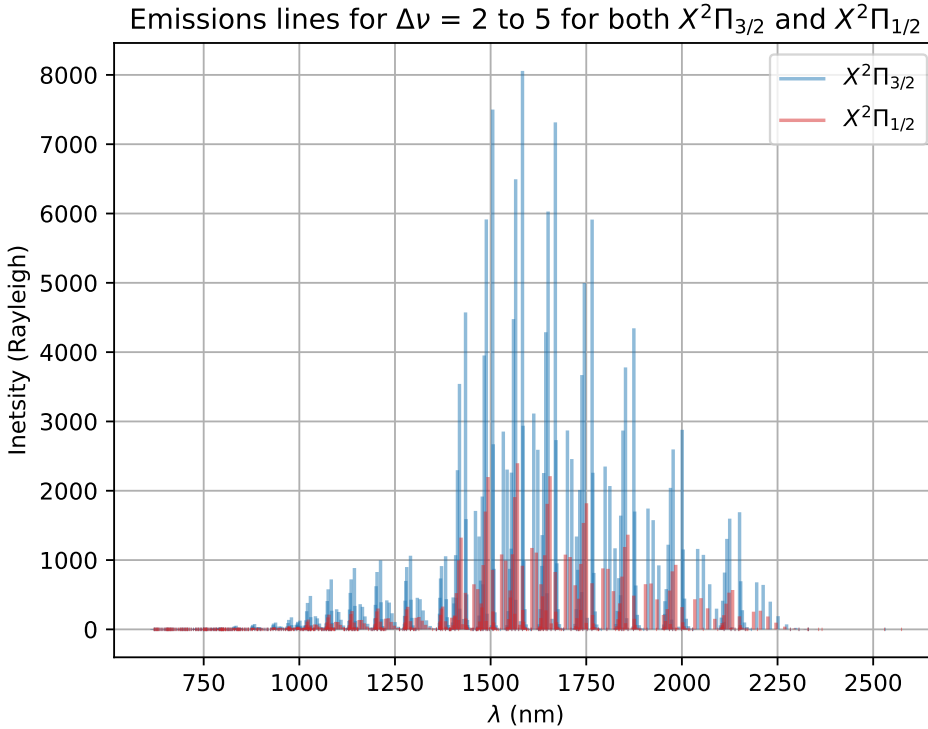


Figure 1: Synthetic Spectrum Overview

At first glance, it is evident that the intensities of the $X^2\Pi_{3/2}$ substate are considerably higher than those of the $X^2\Pi_{1/2}$. To gain a more detailed understanding, the emission lines for each $\Delta\nu$ are separately plotted (Figure 2 and Figure 3) for both substates. A clear trend emerges, showing that the $\Delta\nu = 2$ transitions are the most significant contributors, followed by $\Delta\nu = 3$. In contrast, $\Delta\nu = 4$ and $\Delta\nu = 5$ exhibit notably lower contributions. This pattern holds true for both substates, with $X^2\Pi_{1/2}$ displaying lower intensities overall.

Two separate files, "emission_lines_X2Pi3_2.txt" and "emission_lines_X2Pi1_2.txt", have been created to store the wavelengths, intensities, v' , v'' , J' , and J'' for each substate.

Intensity vs Wavelength for Different Δv for $X^2\Pi_{3/2}$

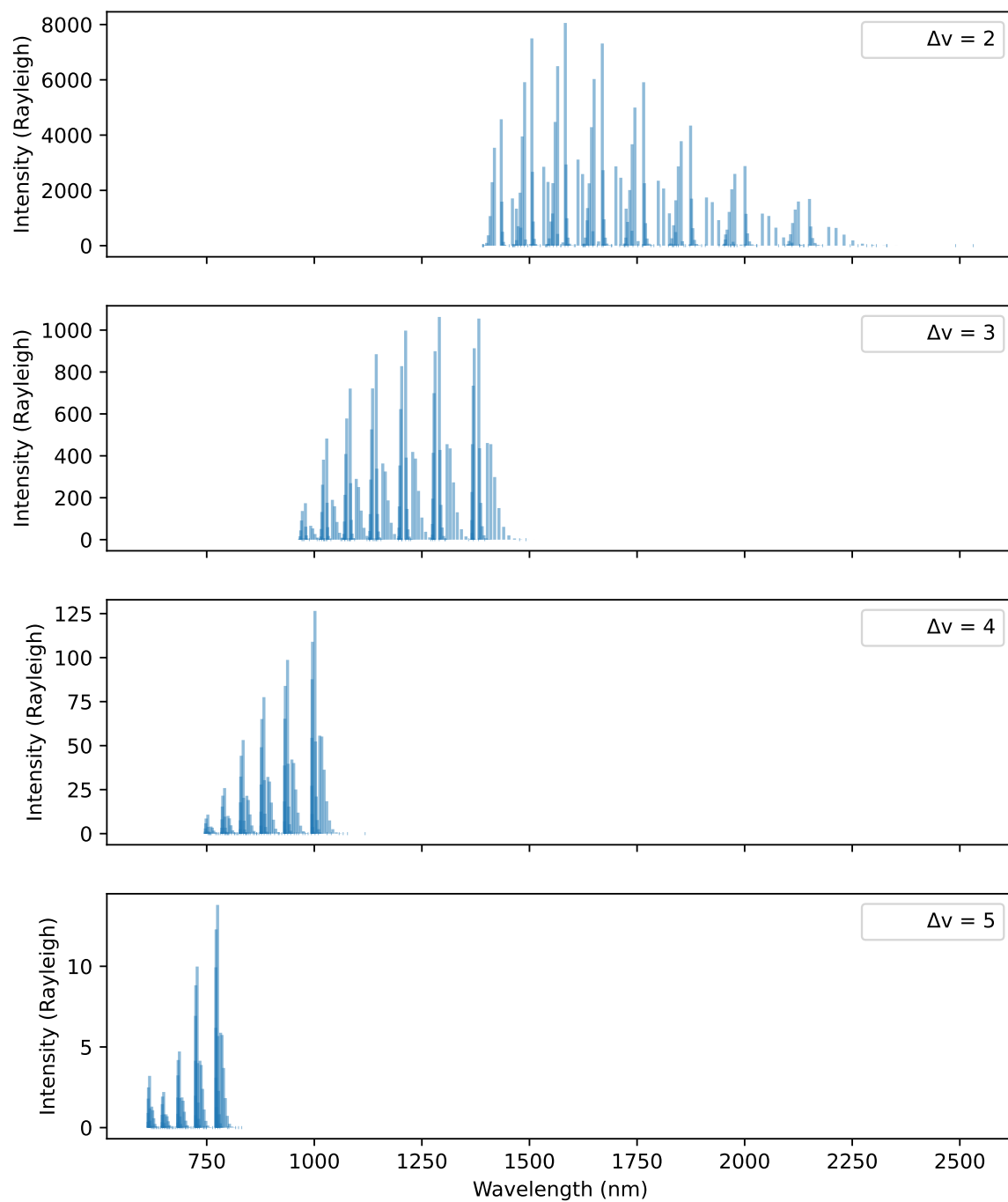


Figure 2: Emission Lines for $X^2\Pi_{3/2}$ Substate

Intensity vs Wavelength for Different Δv for $X^2\Pi_{1/2}$

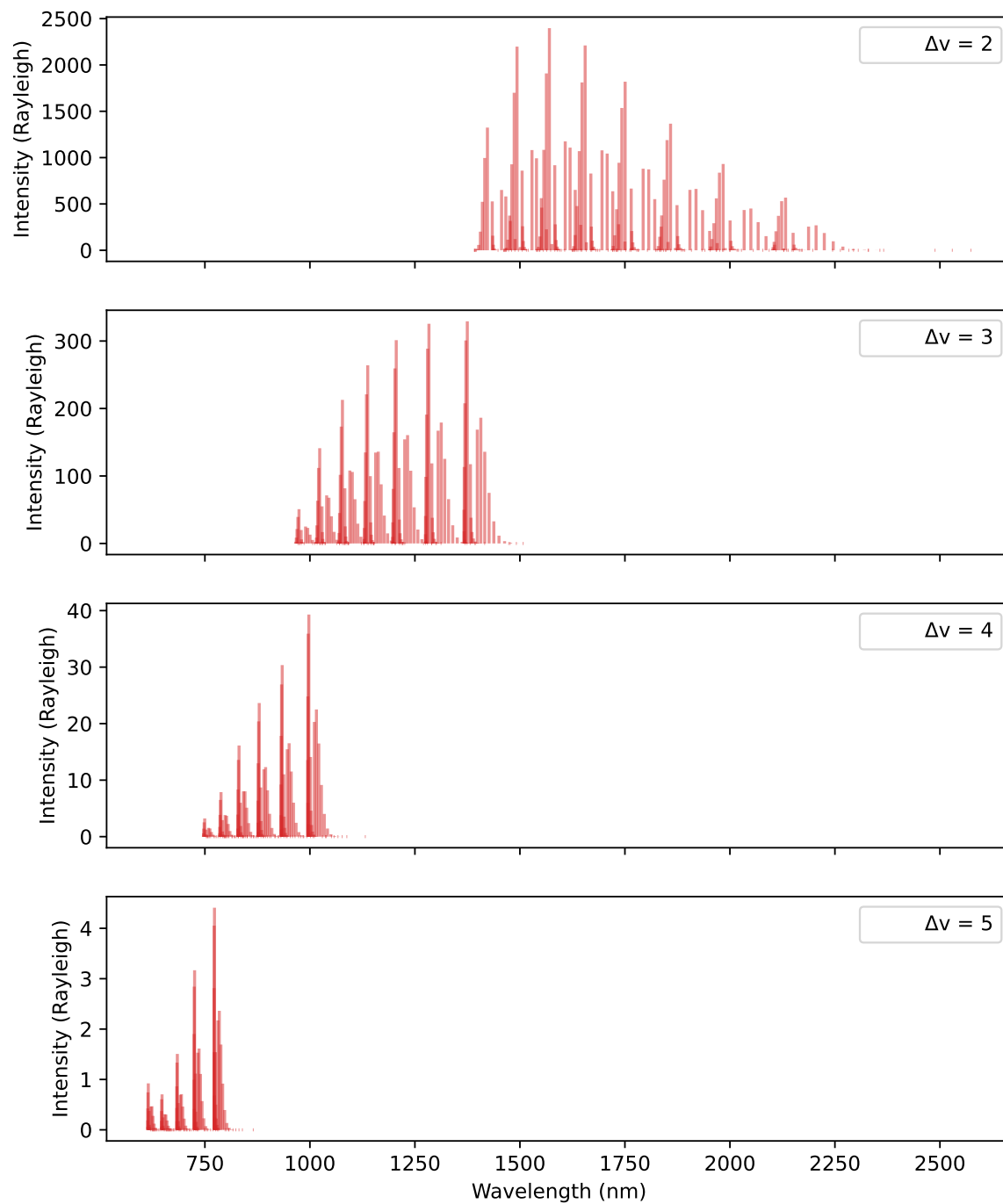


Figure 3: Emission Lines for $X^2\Pi_{1/2}$ Substate

3 Conclusion

We were able to successfully generate a synthetic spectrum for OH radical emissions in the Earth's atmosphere. We observed that the main contributors to the emission lines are those of the $X^2\Pi_{3/2}$ substate, particularly for $\Delta\nu = 2$ and $\Delta\nu = 3$.